

GAURAV GUPTA

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PROFESSIONAL SUMMARY

Software Engineer with 2 years of industry experience building C++ and Python services on AWS and Oracle Cloud for enterprise financial software. Engineered and shipped automated failover system and a cloud migration from AWS to Oracle Cloud. Recently graduated with an MSc in Computer Science from the University of St Andrews, Scotland, UK with coursework focusing on Software Architecture, Distributed Systems, and Software Product Management.

EDUCATION

MSc., Computer Science (16.5/20, Distinction, Dean's List) - University of St Andrews, UK | Sept 2025 - Sept 2026

B.Tech., Computer Science (8.88/10, Distinction) - Dayananda Sagar University, Bengaluru, India | Aug 2019 - May 2023

PROFESSIONAL EXPERIENCE

TALLY SOLUTIONS PVT. LTD. - BENGALURU, INDIA | June 2023 - June 2025

Software Development Engineer II | C++, Python, AWS, Oracle Cloud (OCI)

- Engineered and shipped an **automated failover and self-healing system** on **AWS** for TallyPrime, which helped restore operations for **30+ client systems during an outage** and earned the company's annual **Award as Customer Delight Champion**.
- Migrated **20+ custom Lambda functions across 10+ components** from **AWS** to **Oracle Cloud**, ensuring generic Lambda functionality to eliminate future vendor lock-in.
- Improved core product stability by resolving critical defects in **XML and Math Parsers** used for financial calculations in Tally migration software, enabling enterprise customers to update local desktop apps **without data loss**.
- Delivered key product features, including **file load source identification** to improve **UI/UX** and **placeholder collection components** that allow third-party vendors to customise the Tally UI using **Tally Definition Language (TDL)**, an in-house definition language.

PROJECTS

PRIMAROS: DRUG INTERACTION PLATFORM | **MSC DISSERTATION** | Python, FastAPI, PostgreSQL, Redis

- Architected a **layered clinical decision-support system** integrating three heterogeneous medical data sources (DrugBank, OpenFDA, RxNorm) into a unified domain model of **21,500+ drugs** and **2.9 million interactions**. Built streaming **ingestion pipelines** capable of processing **1.8 GB DrugBank XML** and **8 GB OpenFDA label sets** while keeping memory usage independent of input file size.
- Built a **REST API backend** supporting **trigram-indexed fuzzy drug search** over the ingested data. Added **Redis caching** for recently searched drug interactions to reduce retrieval latency, integrated **Alembic-based database schema migrations**, and supported containerised deployment using **Docker Compose**.

INTERVIEW PREPARATION AGENT | Python, LangGraph, Gemini API, Tavily API | [GitHub](#)

- Built a stateful **LangGraph agent** with three modes: CS course revision, learning, and behavioural coaching. "Revision" dynamically adapts question difficulty using **per-topic confidence scores** persisted in SQLite, while "Coaching" uses **structured LLM extraction** to score responses and provide **STAR-format** feedback. "Learning" tracks competency coverage across sessions, with question banks updated via **Tavily web search**.
- Designed a **CI pipeline with LLM-powered code review** on every push and pull request, generating **severity-graded findings based on predefined criteria** as inline GitHub review comments. Also, built a GitHub workflow that automatically fixes **critical-severity issues identified by the code reviewer agent** on the main branch, with **retry and backoff handling for LLM rate limits**. Well-tested with unit tests using full LLM mocking, executed on every commit.

SKILLS

LANGUAGES & FRAMEWORKS | C++, Python, JavaScript, TypeScript, FastAPI, React, Haskell, Erlang

CLOUD & DEVOPS | AWS (Lambda, EC2, S3), Oracle Cloud Infrastructure (OCI), Docker, CI/CD, GitHub Actions

AI & DATA | LangGraph, LLM Integration, Gemini API, Ollama, PostgreSQL, MongoDB, Redis, SQL

TOOLS & TESTING | pytest, Unit Testing, Git, GitHub, Claude Code, n8n, Postman, Jira, Confluence, Figma

EXTRACURRICULARS AND VOLUNTEERING

- Process, Methodology, and Tools (PMT) trainer and Hackathon Mentor for 10 teams (2024). Tally Solutions Pvt. Ltd.
- C Language Trainer for 70+ students and pre-placement session lead. DSU (University)